

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2022-23

Paper No. STB-301: Probability Theory and Statistical Methods

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) What do you mean by a random sample from a distribution? Distinguish between standard deviation and standard error. Show that the sampling distribution of the sum of two independent Poisson variates is again a Poisson variate. [#1]
- (b) What are the criteria of a good estimator? Explain any two of them with examples. Let $X_1, X_2, \dots, X_n$ be a random sample from a $N(\mu, \sigma^2)$ population, and $T = \sum a_i X_i$ be the statistic based on sample values. Show that:
- (i) $E(T) = \mu$ when $\sum a_i = 1$.
- (ii) $V(T) = \sigma^2 \sum a_i^2$.

[#1]

Question 2. Define null and alternative hypotheses with a suitable example. Explain the concept of Type-I and Type-II errors and bring out their importance in a testing of hypothesis problem. A random sample of size 1 is taken from a Poisson distribution with parameter λ . If $x \geq 2$ is the critical region for testing the hypothesis $H_0 : \lambda = 2$ against $H_1 : \lambda = 1$, find the size and power of the test. [#1]

Question 3.

- (a) Define χ^2 statistic. Describe the χ^2 test for testing independence of attributes. [#1]
- (b) What is the F -distribution? Explain the testing procedure for testing the equality of two variances from two independent normal populations. [#1]

Question 4. State the test of significance for the difference between two means from two independent normal populations when the variances are equal but unknown. Below are given the gains in weights (in kgs) of pigs fed on two diets A and B :

Diet A : 32, 35, 33, 29, 24, 41, 42, 36
Diet B : 35, 36, 39, 37, 38, 25, 26, 30, 24, 20

Test if the two diets differ significantly as regards their effect on increase in weight at 5% level of significance, given that $t_{0.025,16} = 2.12$. [#1]

Question 5.

- (a) Describe the importance of the Central Limit Theorem. State the test of significance for the difference between two proportions in the case of a large sample. [#1]
 - (b) Describe the χ^2 goodness-of-fit test, stating its underlying assumptions. [#1]
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Question 6. Define order statistics along with their various applications. Show that the distribution of the r -th order statistic from a uniform distribution $U(0, 1)$ follows a beta distribution. Also, deduce the distribution of the smallest and largest order statistics. [#1]

Question 7.

- (a) Explain the advantages and disadvantages of non-parametric tests. Describe the one-sample Kolmogorov-Smirnov non-parametric test. [#1]
 - (b) Develop the median non-parametric test, stating the underlying assumptions and the null hypothesis. [#1]
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Question 8. Write short notes on any three of the following:

- i. Estimate and estimator
- ii. Sampling distribution
- iii. Simple and composite hypothesis with examples
- iv. Fisher's t -distribution and its uses
- v. Fisher's z -transformation

[#1]